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$U(1)$ Gauge Field Around a Black Hole

1 Maxwell's Equations in Curved Space: Tree Level

Recall that Maxwell's equations in curved space are written

$$\nabla_\mu F^{\mu\nu} = \partial_\mu F^{\mu\nu} + \Gamma^{\rho}{}_{\mu\rho} F^{\mu\nu} + \Gamma^{\nu}{}_{\rho\mu} F^{\mu\rho} = 4\pi j^\nu \tag{1}$$

where as always $F^{\mu\nu} = \nabla^\mu A^\nu - \nabla^\nu A^\mu = \partial^\mu A^\nu - \partial^\nu A^\mu$. In terms of the electric and magnetic field strengths.

$$F_{\mu\nu} = \begin{bmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & B_3 & -B_2 \\ E_2 & -B_3 & 0 & B_1 \\ E_3 & B_2 & -B_1 & 0 \end{bmatrix} \tag{2}$$

So in terms of the potential, the vacuum maxwell equation become

$$\nabla_\mu F^{\mu\nu} = \Box A^\nu - \nabla_\mu \nabla^\nu A^\mu \tag{3}$$

1.1 Covariant Form

In this paper we will write the maxwell equations for the covariant vector potential A_μ. To get the maxwell equations into a more manageable form let's look at the equation piece by piece starting with the $\Box A_\nu$ term.

$$\begin{aligned}
\Box A_\nu = \nabla_\mu \nabla^\mu A_\nu = g^{\mu\rho} \nabla_\mu \nabla_\rho A_\nu = g^{\mu\rho} \nabla_\mu \left(\partial_\rho A_\nu - \Gamma^\sigma_{\rho\nu} A_\sigma \right) \\
= g^{\mu\rho} \left[\partial_\mu \partial_\rho A_\nu - \partial_\mu \left(\Gamma^\sigma_{\rho\nu} A_\sigma \right) - \Gamma^\lambda_{\nu\mu} \left(\partial_\rho A_\lambda - \Gamma^\sigma_{\rho\lambda} A_\sigma \right) - \Gamma^\lambda_{\rho\mu} \left(\partial_\lambda A_\nu - \Gamma^\sigma_{\lambda\nu} A_\sigma \right) \right] \\
= g^{\mu\rho} \left[\partial_\mu \partial_\rho A_\nu - \partial_\mu \Gamma^\sigma_{\rho\nu} A_\sigma - \Gamma^\sigma_{\rho\nu} \partial_\mu A_\sigma - \Gamma^\lambda_{\nu\mu} \partial_\rho A_\lambda + \Gamma^\lambda_{\nu\mu} \Gamma^\sigma_{\rho\lambda} A_\sigma - \Gamma^\lambda_{\rho\mu} \partial_\lambda A_\nu + \Gamma^\lambda_{\rho\mu} \Gamma^\sigma_{\lambda\nu} A_\sigma \right] \\
= g^{\mu\rho} \left[\partial_\mu \partial_\rho A_\nu - 2\Gamma^\sigma_{\rho\nu} \partial_\mu A_\sigma - \Gamma^\lambda_{\rho\mu} \partial_\lambda A_\nu + \left(-\partial_\mu \Gamma^\sigma_{\rho\nu} + \Gamma^\lambda_{\nu\mu} \Gamma^\sigma_{\rho\lambda} + \Gamma^\lambda_{\rho\mu} \Gamma^\sigma_{\lambda\nu} \right) A_\sigma \right] \\
= g^{\mu\rho} \left[\partial_\mu \partial_\rho A_\nu - 2\Gamma^\sigma_{\rho\nu} \partial_\mu A_\sigma - \Gamma^\lambda_{\rho\mu} \partial_\lambda A_\nu \right] - g^{\mu\rho} \partial_\nu \Gamma^\sigma_{\rho\mu} A_\sigma \\
- g^{\mu\rho} \left(\partial_\mu \Gamma^\sigma_{\rho\nu} - \partial_\nu \Gamma^\sigma_{\rho\mu} - \Gamma^\lambda_{\nu\mu} \Gamma^\sigma_{\rho\lambda} - \Gamma^\lambda_{\rho\mu} \Gamma^\sigma_{\lambda\nu} \right) A_\sigma = g^{\mu\rho} \left[\partial_\mu \partial_\rho A_\nu - 2\Gamma^\sigma_{\rho\nu} \partial_\mu A_\sigma - \Gamma^\lambda_{\rho\mu} \partial_\lambda A_\nu \right] \\
- \partial_\nu (g^{\mu\rho} \Gamma^\sigma_{\rho\mu}) A_\sigma + \Gamma^\sigma_{\rho\mu} \partial_\nu g^{\mu\rho} A_\sigma - g^{\mu\rho} \left(\partial_\mu \Gamma^\sigma_{\rho\nu} - \partial_\nu \Gamma^\sigma_{\rho\mu} - \Gamma^\lambda_{\nu\mu} \Gamma^\sigma_{\rho\lambda} - \Gamma^\lambda_{\rho\mu} \Gamma^\sigma_{\lambda\nu} \right) A_\sigma
\end{aligned} \tag{4}$$

A short detour

$$\begin{aligned}
0 = \partial_\nu \eta^\mu{}_\sigma = \partial_\nu (g^{\mu\lambda} g_{\lambda\sigma}) = g_{\lambda\sigma} \partial_\nu g^{\mu\lambda} + g^{\mu\lambda} \partial_\nu g_{\lambda\sigma} \\
\Rightarrow \quad 0 = g^{\sigma\rho} g_{\lambda\sigma} \partial_\nu g^{\mu\lambda} + g^{\sigma\rho} g^{\mu\lambda} \partial_\nu g_{\lambda\sigma} = \eta^\rho{}_\lambda \partial_\nu g^{\mu\lambda} + g^{\sigma\rho} g^{\mu\lambda} \partial_\nu g_{\lambda\sigma} \\
= \partial_\nu g^{\mu\rho} + g^{\sigma\rho} g^{\mu\lambda} \partial_\nu g_{\lambda\sigma} \quad \Rightarrow \quad \partial_\nu g^{\mu\rho} = -g^{\omega\rho} g^{\mu\lambda} \partial_\nu g_{\lambda\omega}
\end{aligned} \tag{5}$$

continuing

$$\begin{aligned}
\Box A_\nu = g^{\mu\rho}\left[\partial_\mu\partial_\rho A_\nu - 2\Gamma^\sigma_{\rho\nu}\partial_\mu A_\sigma - \Gamma^\lambda_{\rho\mu}\partial_\lambda A_\nu\right] \\
-\partial_\nu(g^{\mu\rho}\Gamma^\sigma_{\rho\mu})A_\sigma - \Gamma^\sigma_{\rho\mu}g^{\omega\rho}g^{\mu\lambda}\partial_\nu g_{\lambda\omega}A_\sigma - g^{\mu\rho}\left(\partial_\mu\Gamma^\sigma_{\rho\nu} - \partial_\nu\Gamma^\sigma_{\rho\mu} - \Gamma^\lambda_{\nu\mu}\Gamma^\sigma_{\rho\lambda} - \Gamma^\lambda_{\rho\mu}\Gamma^\sigma_{\lambda\nu}\right)A_\sigma \\
= g^{\mu\rho}\left[\partial_\mu\partial_\rho A_\nu - 2\Gamma^\sigma_{\rho\nu}\partial_\mu A_\sigma - \Gamma^\lambda_{\rho\mu}\partial_\lambda A_\nu\right] \\
-\partial_\nu(g^{\mu\rho}\Gamma^\sigma_{\rho\mu})A_\sigma - \Gamma^\sigma_{\lambda\rho}g^{\omega\lambda}g^{\rho\mu}\left(\partial_\nu g_{\mu\omega} + \partial_\mu g_{\nu\omega} - \partial_\omega g_{\mu\nu}\right)A_\sigma \\
-g^{\mu\rho}\left(\partial_\mu\Gamma^\sigma_{\rho\nu} - \partial_\nu\Gamma^\sigma_{\rho\mu} - \Gamma^\lambda_{\nu\mu}\Gamma^\sigma_{\rho\lambda} - \Gamma^\lambda_{\rho\mu}\Gamma^\sigma_{\lambda\nu}\right)A_\sigma \\
= g^{\mu\rho}\left[\partial_\mu\partial_\rho A_\nu - 2\Gamma^\sigma_{\rho\nu}\partial_\mu A_\sigma - \Gamma^\lambda_{\rho\mu}\partial_\lambda A_\nu\right] \\
-\partial_\nu(g^{\mu\rho}\Gamma^\sigma_{\rho\mu})A_\sigma - 2\Gamma^\sigma_{\lambda\rho}\Gamma^\lambda_{\mu\nu}g^{\rho\mu}A_\sigma - g^{\mu\rho}\left(\partial_\mu\Gamma^\sigma_{\rho\nu} - \partial_\nu\Gamma^\sigma_{\rho\mu} - \Gamma^\lambda_{\nu\mu}\Gamma^\sigma_{\rho\lambda} - \Gamma^\lambda_{\rho\mu}\Gamma^\sigma_{\lambda\nu}\right)A_\sigma \\
= g^{\mu\rho}\left[\partial_\mu\partial_\rho A_\nu - 2\Gamma^\sigma_{\rho\nu}\partial_\mu A_\sigma - \Gamma^\lambda_{\rho\mu}\partial_\lambda A_\nu\right] \\
-\partial_\nu(g^{\mu\rho}\Gamma^\sigma_{\rho\mu})A_\sigma - g^{\mu\rho}\left(\partial_\mu\Gamma^\sigma_{\nu\rho} - \partial_\nu\Gamma^\sigma_{\mu\rho} + \Gamma^\sigma_{\rho\lambda}\Gamma^\lambda_{\nu\mu} - \Gamma^\sigma_{\nu\lambda}\Gamma^\lambda_{\rho\mu}\right)A_\sigma \\
= g^{\mu\rho}\left[\partial_\mu\partial_\rho A_\nu - 2\Gamma^\sigma_{\rho\nu}\partial_\mu A_\sigma - \Gamma^\lambda_{\rho\mu}\partial_\lambda A_\nu\right] \\
-\partial_\nu(g^{\mu\rho}\Gamma^\sigma_{\rho\mu})A_\sigma - g^{\mu\rho}\left(\partial_\mu\Gamma^\sigma_{\nu\rho} - \partial_\nu\Gamma^\sigma_{\mu\rho} + \Gamma^\sigma_{\mu\lambda}\Gamma^\lambda_{\nu\rho} - \Gamma^\sigma_{\nu\lambda}\Gamma^\lambda_{\mu\rho}\right)A_\sigma \\
= g^{\mu\rho}\left[\partial_\mu\partial_\rho A_\nu - 2\Gamma^\sigma_{\rho\nu}\partial_\mu A_\sigma - \Gamma^\lambda_{\rho\mu}\partial_\lambda A_\nu\right] - \partial_\nu(g^{\mu\rho}\Gamma^\sigma_{\rho\mu})A_\sigma - g^{\mu\rho}R^\sigma{}_{\rho\mu\nu}A_\sigma
\end{aligned} \tag{6}$$

Using the symmetry properties of the curvature tensor

$$g^{\mu\rho}R^\sigma{}_{\rho\mu\nu} = g^{\sigma\lambda}g^{\mu\rho}R_{\lambda\rho\mu\nu} = -g^{\sigma\lambda}g^{\mu\rho}R_{\rho\lambda\mu\nu} = -g^{\sigma\lambda}R_{\lambda\nu} \tag{7}$$

Thus, $\Box A_\nu$ becomes:

$$\Box A_\nu = LA_\nu - 2g^{\mu\rho}\Gamma^\sigma_{\rho\nu}\partial_\mu A_\sigma - \partial_\nu(g^{\mu\rho}\Gamma^\sigma_{\rho\mu})A_\sigma + g^{\sigma\mu}R_{\mu\nu}A_\sigma \tag{8}$$

where L is the scalar D'Alembert operator defined by

$$LA_\nu = \frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}g^{\mu\rho}\partial_\rho A_\nu) = g^{\mu\rho}\left[\partial_\mu\partial_\rho A_\nu - \Gamma^\lambda_{\rho\mu}\partial_\lambda A_\nu\right] \tag{9}$$

Thus, Maxwell's equation becomes

$$\begin{aligned}
\Box A_\nu - \nabla_\mu\nabla_\nu A^\mu = \Box A_\nu - \nabla_\nu\nabla_\mu A^\mu - R_{\nu\mu}A^\mu \\
= LA_\nu - 2g^{\mu\rho}\Gamma^\sigma_{\rho\nu}\partial_\mu A_\sigma - \partial_\nu(g^{\mu\rho}\Gamma^\sigma_{\rho\mu})A_\sigma + g^{\sigma\mu}R_{\mu\nu}A_\sigma - \nabla_\nu\nabla_\mu A^\mu - R_{\nu\mu}A^\mu \\
= LA_\nu - 2g^{\mu\rho}\Gamma^\sigma_{\rho\nu}\partial_\mu A_\sigma - \partial_\nu(g^{\mu\rho}\Gamma^\sigma_{\rho\mu})A_\sigma + R_{\mu\nu}A^\mu - \nabla_\nu\nabla_\mu A^\mu - R_{\mu\nu}A^\mu \\
= LA_\nu - 2g^{\mu\rho}\Gamma^\sigma_{\rho\nu}\partial_\mu A_\sigma - \partial_\nu(g^{\mu\rho}\Gamma^\sigma_{\rho\mu})A_\sigma - \nabla_\nu\nabla_\mu A^\mu \\
\Rightarrow \quad LA_\nu = 2g^{\mu\rho}\Gamma^\sigma_{\rho\nu}\partial_\mu A_\sigma + \partial_\nu(g^{\mu\rho}\Gamma^\sigma_{\rho\mu})A_\sigma + \partial_\nu(\nabla_\mu A^\mu)
\end{aligned} \tag{10}$$

1.2 Schwarzschild Space with Stereographic Coordinates

The schwarzschild metric is given by

$$ds^2 = (1 - \frac{2M}{r})dt^2 - (1 - \frac{2M}{r})^{-1}dr^2 - r^2d\theta^2 - r^2\sin^2\theta d\varphi^2 \tag{11}$$

It is more covinent to use stereographic coordinates $(z, \bar{z})$ in place of the angular coordinates (θ, φ). The definition of the stereographic coordinates $(z, \bar{z})$ in terms of (θ, φ) is given by

$$z = \frac{\sin\theta e^{i\varphi}}{1-\cos\theta}, \quad \bar{z} = \frac{\sin\theta e^{-i\varphi}}{1-\cos\theta} \tag{12}$$

The metric in terms of these stereographic coordinates is

$$ds^2 = (1-\frac{2M}{r})dt^2 - (1-\frac{2M}{r})^{-1}dr^2 - \frac{4r^2}{(1+z\bar{z})^2}dzd\bar{z} \tag{13}$$

The Christoffel terms are then

$$\begin{gathered}
g^{\mu\rho}\Gamma^{\sigma}_{\rho 0} = \Gamma^{\sigma\mu}{}_{0} = \begin{bmatrix} 0 & -\frac{M}{r^2} & 0 & 0 \\ \frac{M}{r^2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\
\Gamma^{\sigma\mu}{}_{1} = \begin{bmatrix} \frac{M}{r^2(1-\frac{2M}{r})^2} & 0 & 0 & 0 \\ 0 & \frac{M}{r^2} & 0 & 0 \\ 0 & 0 & 0 & -\frac{(1+z\bar{z})^2}{2r^3} \\ 0 & 0 & -\frac{(1+z\bar{z})^2}{2r^3} & 0 \end{bmatrix} \\
\Gamma^{\sigma\mu}{}_{2} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{r}(1-\frac{2M}{r}) & 0 \\ 0 & -\frac{1}{r}(1-\frac{2M}{r}) & 0 & \frac{\bar{z}(1+z\bar{z})}{r^2} \\ 0 & 0 & 0 & 0 \end{bmatrix} \\
\Gamma^{\sigma\mu}{}_{3} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{r}(1-\frac{2M}{r}) \\ 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{r}(1-\frac{2M}{r}) & \frac{z(1+z\bar{z})}{r^2} & 0 \end{bmatrix}
\end{gathered} \tag{14}$$

and

$$\begin{gathered}
g^{\mu\nu}\Gamma^{\rho}_{\mu\nu} = \left\{0, \frac{2(r-M)}{r^2}, 0, 0\right\} \\
\partial_0\left(g^{\mu\nu}\Gamma^{\rho}_{\mu\nu}\right) = \{0,0,0,0\}, \quad \partial_1\left(g^{\mu\nu}\Gamma^{\rho}_{\mu\nu}\right) = \left\{0, -\frac{2}{r^2}(1-\frac{2M}{r}), 0, 0\right\}, \\
\partial_2\left(g^{\mu\nu}\Gamma^{\rho}_{\mu\nu}\right) = \{0,0,0,0\}, \quad \partial_3\left(g^{\mu\nu}\Gamma^{\rho}_{\mu\nu}\right) = \{0,0,0,0\}
\end{gathered} \tag{15}$$

$$\begin{aligned}
\Gamma^{\sigma\mu}{}_{0}\partial_\mu A_\sigma &= \frac{M}{r^2}\left(\partial_0 A_1 - \partial_1 A_0\right), \\
\Gamma^{\sigma\mu}{}_{1}\partial_\mu A_\sigma &= \frac{M}{r^2(1-\frac{2M}{r})^2}\partial_0 A_0 + \frac{M}{r^2}\partial_1 A_1 - \frac{(1+z\bar{z})^2}{2r^3}(\partial_2 A_3 + \partial_3 A_2), \\
\Gamma^{\sigma\mu}{}_{2}\partial_\mu A_\sigma &= -\frac{1}{r}(1-\frac{2M}{r})(\partial_1 A_2 - \partial_2 A_1) + \frac{\bar{z}(1+z\bar{z})}{r^2}\partial_3 A_2, \\
\Gamma^{\sigma\mu}{}_{3}\partial_\mu A_\sigma &= -\frac{1}{r}(1-\frac{2M}{r})(\partial_1 A_3 - \partial_3 A_1) + \frac{z(1+z\bar{z})}{r^2}\partial_2 A_3
\end{aligned} \tag{16}$$

$$\partial_0\left(g^{\mu\nu}\Gamma^{\rho}_{\mu\nu}\right)A_\rho = 0, \quad \partial_1\left(g^{\mu\nu}\Gamma^{\rho}_{\mu\nu}\right) = -\frac{2}{r^2}(1-\frac{2M}{r})A_1,$$
$$\partial_2\left(g^{\mu\nu}\Gamma^{\rho}_{\mu\nu}\right)A_\rho = 0, \quad \partial_3\left(g^{\mu\nu}\Gamma^{\rho}_{\mu\nu}\right)A_\rho = 0 \tag{17}$$

So the equations of motion

$$LA_0 = \frac{2M}{r^2}\left(\partial_0 A_1 - \partial_1 A_0\right) + \partial_0 \nabla_\mu A^\mu \tag{18}$$
$$LA_1 = \frac{2M}{r^2(1-\frac{2M}{r})^2}\partial_0 A_0 + \frac{2M}{r^2}\partial_1 A_1$$
$$-\frac{(1+z\bar{z})^2}{r^3}(\partial_2 A_3 + \partial_3 A_2) - \frac{2}{r^2}(1-\frac{2M}{r})A_1 + \partial_1 \nabla_\mu A^\mu \tag{19}$$
$$LA_2 = -\frac{2}{r}(1-\frac{2M}{r})(\partial_1 A_2 - \partial_2 A_1) + \frac{2\bar{z}(1+z\bar{z})}{r^2}\partial_3 A_2 + \partial_2 \nabla_\mu A^\mu \tag{20}$$
$$LA_3 = -\frac{2}{r}(1-\frac{2M}{r})(\partial_1 A_3 - \partial_3 A_1) + \frac{2z(1+z\bar{z})}{r^2}\partial_2 A_3 + \partial_3 \nabla_\mu A^\mu \tag{21}$$

Expand out $\nabla_\mu A^\mu$

$$\nabla_\mu A^\mu = g^{\mu\nu}\nabla_\mu A_\nu = g^{\mu\nu}\left(\partial_\mu A_\nu - \Gamma^{\lambda}_{\mu\nu}A_\lambda\right)$$
$$= (1-\frac{2M}{r})^{-1}\partial_0 A_0 - (1-\frac{2M}{r})\partial_1 A_1 - \frac{(1+z\bar{z})^2}{2r^2}(\partial_2 A_3 + \partial_3 A_2) - \frac{2(r-M)}{r^2}A_1$$
$$= (1-\frac{2M}{r})^{-1}\partial_0 A_0 - (1-\frac{2M}{r})\frac{1}{r^2}\partial_1(r^2 A_1) - \frac{2M}{r^2}A_1 - \frac{(1+z\bar{z})^2}{2r^2}(\partial_2 A_3 + \partial_3 A_2) \tag{22}$$

Fix the gauge so that $A_0 = 0$, then equation 18 becomes

$$0 = \partial_0\left[\frac{2M}{r^2}A_1 - (1-\frac{2M}{r})\partial_1 A_1 - \frac{(1+z\bar{z})^2}{2r^2}(\partial_2 A_3 + \partial_3 A_2) - \frac{2(r-M)}{r^2}A_1\right]$$
$$= -\partial_0\left[(1-\frac{2M}{r})\partial_1 A_1 + \frac{(1+z\bar{z})^2}{2r^2}(\partial_2 A_3 + \partial_3 A_2) + \frac{2}{r}(1-\frac{2M}{r})A_1\right]$$
$$= -\partial_0\left[(1-\frac{2M}{r})\frac{1}{r^2}\partial_1(r^2 A_1) + \frac{(1+z\bar{z})^2}{2r^2}(\partial_2 A_3 + \partial_3 A_2)\right] \tag{23}$$

Thus $\nabla_\mu A^\mu$ becomes in this gauge

$$\nabla_\mu A^\mu = -\frac{2M}{r^2}A_1 \tag{24}$$

Thus equation (19)

$$
\begin{aligned}
LA_1 &= \frac{1}{r}\left[-(1-\frac{2M}{r})\frac{1}{r^2}\partial_1(r^2A_1)+\partial_1A_1-\frac{(1+z\bar{z})^2}{r^2}(\partial_2A_3+\partial_3A_2)+r\partial_1\nabla_\mu A^\mu\right]\\
\Rightarrow \quad LA_1 &= \frac{1}{r}\left[(1-\frac{2M}{r})\frac{1}{r^2}\partial_1(r^2A_1)+\partial_1A_1-2Mr\partial_1\left(\frac{A_1}{r^2}\right)\right]\\
\Rightarrow \quad LA_1 &= \frac{1}{r}\left[(1-\frac{2M}{r})\frac{1}{r^2}\partial_1(r^2A_1)+\partial_1A_1+\frac{4M}{r^2}A_1-\frac{2M}{r}\partial_1A_1\right]\\
\Rightarrow \quad LA_1 &= \frac{1}{r}\left[2(1-\frac{2M}{r})\partial_1A_1+\frac{2}{r}(1-\frac{2M}{r})A_1+\frac{4M}{r^2}A_1\right]\\
\Rightarrow \quad LA_1 &= \frac{2}{r}\left[(1-\frac{2M}{r})\partial_1A_1+\frac{1}{r}(1-\frac{2M}{r})A_1+\frac{2M}{r^2}A_1\right]\\
\Rightarrow \quad LA_1 &= \frac{2}{r}\left[(1-\frac{2M}{r})\partial_1A_1+\frac{1}{r}\partial_1\left(r(1-\frac{2M}{r})\right)A_1\right]\\
\Rightarrow \quad LA_r &= \frac{2}{r^2}\partial_r\left[r(1-\frac{2M}{r})A_r\right]
\end{aligned}
\tag{25}
$$

where $A_r = A_1$ and $\partial_r = \partial_1$. Equation (20)

$$
\begin{aligned}
LA_2 &= -\frac{2}{r}(1-\frac{2M}{r})(\partial_1A_2-\partial_2A_1)+\frac{2\bar{z}(1+z\bar{z})}{r^2}\partial_3A_2-\frac{2M}{r^2}\partial_2A_1\\
\Rightarrow \quad \left(L-\frac{2\bar{z}(1+z\bar{z})}{r^2}\partial_{\bar{z}}\right)A_z &= -\frac{2}{r}(1-\frac{2M}{r})\partial_rA_z+\frac{2}{r}(1-\frac{3M}{r})\partial_zA_r
\end{aligned}
\tag{26}
$$

where, $A_z = A_2$, $\partial_z = \partial_2$ and $\partial_{\bar{z}} = \partial_3$. Equation (21) becomes

$$
\begin{aligned}
LA_3 &= -\frac{2}{r}(1-\frac{2M}{r})(\partial_1A_3-\partial_3A_1)+\frac{2z(1+z\bar{z})}{r^2}\partial_2A_3-\frac{2M}{r^2}\partial_3A_1\\
\Rightarrow \quad \left(L-\frac{2z(1+z\bar{z})}{r^2}\partial_z\right)A_{\bar{z}} &= -\frac{2}{r}(1-\frac{2M}{r})\partial_rA_{\bar{z}}+\frac{2}{r}(1-\frac{3M}{r})\partial_{\bar{z}}A_r
\end{aligned}
\tag{27}
$$

where $A_{\bar{z}} = A_3$. Let's concentrate on the left hand side of (26)

$$
\left(L-\frac{2\bar{z}(1+z\bar{z})}{r^2}\partial_{\bar{z}}\right)\partial_z\psi \tag{28}
$$

Looking in particular at the $L\partial_z\psi$ piece

$$
\begin{aligned}
L\partial_z\psi &= \frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}g^{\mu\nu}\partial_\nu\partial_z\psi) = \frac{1}{\sqrt{-g}}\partial_\mu\left[\partial_z(\sqrt{-g}g^{\mu\nu}\partial_\nu\psi) - \partial_z(\sqrt{-g}g^{\mu\nu})\partial_\nu\psi\right] \\
&= \frac{1}{\sqrt{-g}}\partial_\mu\partial_z(\sqrt{-g}g^{\mu\nu}\partial_\nu\psi) - \frac{1}{\sqrt{-g}}\partial_\mu\left[\partial_z(\sqrt{-g}g^{\mu\nu})\partial_\nu\psi\right] \\
&= \partial_z\left[\frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}g^{\mu\nu}\partial_\nu\psi)\right] - \partial_z\left[\frac{1}{\sqrt{-g}}\right]\partial_\mu(\sqrt{-g}g^{\mu\nu}\partial_\nu\psi) - \frac{1}{\sqrt{-g}}\partial_\mu\left[\partial_z(\sqrt{-g}g^{\mu\nu})\partial_\nu\psi\right] \\
&= \partial_z L\psi + \frac{1}{-g}\partial_z\left[\sqrt{-g}\right]\partial_\mu(\sqrt{-g}g^{\mu\nu}\partial_\nu\psi) - \frac{1}{\sqrt{-g}}\partial_\mu\left[\partial_z(\sqrt{-g}g^{\mu\nu})\partial_\nu\psi\right] \\
&= \partial_z L\psi + \frac{1}{-g}\partial_z\left[\sqrt{-g}\right]\partial_\mu(\sqrt{-g}g^{\mu\nu})\partial_\nu\psi + \frac{1}{\sqrt{-g}}\partial_z\left[\sqrt{-g}\right]g^{\mu\nu}\partial_\mu\partial_\nu\psi \\
&\quad - \frac{1}{\sqrt{-g}}\partial_\mu\partial_z(\sqrt{-g}g^{\mu\nu})\partial_\nu\psi - \frac{1}{\sqrt{-g}}\partial_z(\sqrt{-g}g^{\mu\nu})\partial_\mu\partial_\nu\psi \\
&= \partial_z L\psi + \frac{1}{\sqrt{-g}}\left[\frac{\partial_z\sqrt{-g}}{\sqrt{-g}}\partial_\mu(\sqrt{-g}g^{\mu\nu}) - \partial_\mu\partial_z(\sqrt{-g}g^{\mu\nu})\right]\partial_\nu\psi - \partial_z g^{\mu\nu}\partial_\mu\partial_\nu\psi \\
&= \partial_z L\psi - \frac{1}{\sqrt{-g}}\left[(\partial_z\sqrt{-g})\Gamma^\nu_{\lambda\rho}g^{\lambda\rho} - \partial_z(\sqrt{-g}\Gamma^\nu_{\lambda\rho}g^{\lambda\rho})\right]\partial_\nu\psi - \partial_z g^{\mu\nu}\partial_\mu\partial_\nu\psi \\
&= \partial_z L\psi - \partial_z g^{\mu\nu}\partial_\mu\partial_\nu\psi + \partial_z(\Gamma^\lambda_{\mu\nu}g^{\mu\nu})\partial_\lambda\psi
\end{aligned}
\tag{29}
$$

Note that $\partial_z(\Gamma^\lambda_{\mu\nu}g^{\mu\nu}) = 0$ as we showed earlier, and also

$$
\partial_z g^{\mu\nu} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\bar{z}(1+z\bar{z})}{r^2} \\ 0 & 0 & -\frac{\bar{z}(1+z\bar{z})}{r^2} & 0 \end{bmatrix}
\tag{30}
$$

$$
L\partial_z\psi = \partial_z L\psi + \frac{2\bar{z}(1+z\bar{z})}{r^2}\partial_{\bar{z}}\partial_z\psi \quad \Rightarrow \quad \left(L - \frac{2\bar{z}(1+z\bar{z})}{r^2}\partial_{\bar{z}}\right)\partial_z\psi = \partial_z L\psi
\tag{31}
$$

Looking at $L\partial_{\bar{z}}\psi$:

$$
L\partial_{\bar{z}}\psi = \partial_{\bar{z}} L\psi - \partial_{\bar{z}} g^{\mu\nu}\partial_\mu\partial_\nu\psi + \partial_{\bar{z}}(\Gamma^\lambda_{\mu\nu}g^{\mu\nu})\partial_\lambda\psi
\tag{32}
$$

Similarly,

$$
L\partial_{\bar{z}}\psi = \partial_{\bar{z}} L\psi + \frac{2z(1+z\bar{z})}{r^2}\partial_z\partial_{\bar{z}}\psi \quad \Rightarrow \quad \left(L - \frac{2z(1+z\bar{z})}{r^2}\partial_z\right)\partial_{\bar{z}}\psi = \partial_{\bar{z}} L\psi
\tag{33}
$$

The equations of motion are

$$
LA_r = \frac{2}{r^2}\partial_r\left[r(1-\frac{2M}{r})A_r\right]
\tag{34}
$$

$$
\left(L - \frac{2\bar{z}(1+z\bar{z})}{r^2}\partial_{\bar{z}}\right)A_z = -\frac{2}{r}(1-\frac{2M}{r})\partial_r A_z + \frac{2}{r}(1-\frac{3M}{r})\partial_z A_r
\tag{35}
$$

$$
\left(L - \frac{2z(1+z\bar{z})}{r^2}\partial_z\right)A_{\bar{z}} = -\frac{2}{r}(1-\frac{2M}{r})\partial_r A_{\bar{z}} + \frac{2}{r}(1-\frac{3M}{r})\partial_{\bar{z}} A_r
\tag{36}
$$

and the constraint is

$$(1-\frac{2M}{r})\partial_r(r^2A_r)+\frac{(1+z\bar{z})^2}{2}(\partial_zA_{\bar{z}}+\partial_{\bar{z}}A_z)=0 \tag{37}$$

1.3 From Vector to Scalar Equations

1.3.1 First Solution

Suppose $A_z=\partial_z\phi_1$, where ϕ_1 is a scalar function, then equation (35) becomes

$$\begin{aligned}\partial_zL\phi_1 &= -\frac{2}{r}(1-\frac{2M}{r})\partial_r\partial_z\phi_1+\frac{2}{r}(1-\frac{3M}{r})\partial_zA_r\\ \Rightarrow\quad \partial_z&\left(L\phi_1+\frac{2}{r}(1-\frac{2M}{r})\partial_r\phi_1-\frac{2}{r}(1-\frac{3M}{r})A_r\right)=0\end{aligned} \tag{38}$$

and if $A_{\bar{z}}=\partial_{\bar{z}}\phi_2$ then equation (36) becomes

$$\begin{aligned}\partial_{\bar{z}}L\phi_2 &= -\frac{2}{r}(1-\frac{2M}{r})\partial_r\partial_{\bar{z}}\phi_2+\frac{2}{r}(1-\frac{3M}{r})\partial_{\bar{z}}A_r\\ \Rightarrow\quad \partial_{\bar{z}}&\left(L\phi_2+\frac{2}{r}(1-\frac{2M}{r})\partial_r\phi_2-\frac{2}{r}(1-\frac{3M}{r})A_r\right)=0\end{aligned} \tag{39}$$

Integrating and subtracting these two equations yields

$$L(\phi_1-\phi_2)+\frac{2}{r}(1-\frac{2M}{r})\partial_r(\phi_1-\phi_2)=F(t,r,\bar{z})-G(t,r,z) \tag{40}$$

where F and G are arbitrary functions that result from integration. The constraint equation (37) becomes

$$(1-\frac{2M}{r})\partial_r(r^2A_r)+\frac{(1+z\bar{z})^2}{2}\partial_z\partial_{\bar{z}}(\phi_1+\phi_2)=0 \tag{41}$$

What if $\phi_1=-\phi_2=\Phi$? Then the constraint becomes

$$(1-\frac{2M}{r})\partial_r(r^2A_r)=0\quad\Rightarrow\quad r^2A_r=H(t,z,\bar{z}) \tag{42}$$

The A_r equation of motion becomes

$$\frac{1}{1-\frac{2M}{r}}\partial_t^2H=\frac{(1+z\bar{z})^2}{r^2}\partial_z\partial_{\bar{z}}H \tag{43}$$

If H is nonzero

$$\frac{\partial_t^2H}{\partial_z\partial_{\bar{z}}H}=\frac{(1+z\bar{z})^2}{r^2(\frac{1}{1-\frac{2M}{r}})} \tag{44}$$

The left hand side is independent of r, thus we arrive at a contradiction, thus $H = 0$. and equation (41)

$$L\Phi + \frac{2}{r}(1 - \frac{2M}{r})\partial_r \Phi = \frac{F(t,r,\bar{z}) - G(t,r,z)}{2} \tag{45}$$

Let's just say for now that the right hand side is zero then

$$L\Phi = -\frac{2}{r}(1 - \frac{2M}{r})\partial_r \Phi \tag{46}$$

This can be expressed as

$$\begin{aligned}
&g^{tt}\partial_t^2\Phi + \frac{1}{r^2}\partial_r(r^2 g^{rr}\partial_r\Phi) + (1+z\bar{z})^2\partial_z((1+z\bar{z})^{-2}g^{z\bar{z}}\partial_{\bar{z}}\Phi) \\
&+(1+z\bar{z})^2\partial_{\bar{z}}((1+z\bar{z})^{-2}g^{\bar{z}z}\partial_z\Phi) = -\frac{2}{r}(1-\frac{2M}{r})\partial_r\Phi \\
\Rightarrow \quad &(1-\frac{2M}{r})^{-1}\partial_t^2\Phi - \frac{1}{r^2}\partial_r\left(r^2(1-\frac{2M}{r})\partial_r\Phi\right) - \frac{(1+z\bar{z})^2}{r^2}\partial_z\partial_{\bar{z}}\Phi = -\frac{2}{r}(1-\frac{2M}{r})\partial_r\Phi \\
\Rightarrow \quad &(1-\frac{2M}{r})^{-1}\partial_t^2\Phi - \frac{1}{r^2}\partial_r\left(r^2(1-\frac{2M}{r})\partial_r\Phi\right) \\
&-\frac{1}{r^2}\left(\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Phi}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2\Phi}{\partial\varphi^2}\right) = -\frac{2}{r}(1-\frac{2M}{r})\partial_r\Phi
\end{aligned} \tag{47}$$

Where we have used the fact that:

$$(1+z\bar{z})^2\partial_z\partial_{\bar{z}}\Phi = \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Phi}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2\Phi}{\partial\varphi^2} \tag{48}$$

The proof is in the appendix. Let's suppose that $\Phi(t,r,\theta,\varphi) = f(r)Y_l^m(\theta,\varphi)e^{i\omega t}$. Thus

$$\begin{aligned}
&-(1-\frac{2M}{r})^{-1}\omega^2 f(r) - \frac{1}{r^2}\partial_r\left(r^2(1-\frac{2M}{r})\partial_r f(r)\right) + \frac{l(l+1)}{r^2}f(r) = -\frac{2}{r}(1-\frac{2M}{r})\partial_r f(r) \\
\Rightarrow \quad &-(1-\frac{2M}{r})^{-1}\omega^2 f(r) - \frac{1}{r^2}\partial_r\left(r^2(1-\frac{2M}{r})\partial_r f(r)\right) + \frac{2}{r}(1-\frac{2M}{r})\partial_r f(r) \\
&= -\frac{l(l+1)}{r^2}f(r) \quad \Rightarrow \quad -(1-\frac{2M}{r})^{-1}\omega^2 f(r) - \partial_r\left((1-\frac{2M}{r})\partial_r f(r)\right) \\
&-\frac{2}{r}(1-\frac{2M}{r})\partial_r f(r) + \frac{2}{r}(1-\frac{2M}{r})\partial_r f(r) = -\frac{l(l+1)}{r^2}f(r) \\
\Rightarrow \quad &(1-\frac{2M}{r})^{-1}\omega^2 f(r) + \partial_r\left((1-\frac{2M}{r})\partial_r f(r)\right) = \frac{l(l+1)}{r^2}f(r)
\end{aligned} \tag{49}$$

Define the tortoise coordinate as

$$r^* = r + 2M\log\left|\frac{r}{2M} - 1\right| \tag{50}$$

So

$$\frac{dr^*}{dr} = (1 - \frac{2M}{r})^{-1} \tag{51}$$

and therefore

$$(1-\frac{2M}{r})^{-1}\omega^2 f(r) + (1-\frac{2M}{r})^{-1}\partial_{r^*}\left(\partial_{r^*} f(r)\right) = \frac{l(l+1)}{r^2} f(r) \tag{52}$$

Cleaning things up a bit

$$\left(\partial_{r^*}^2 + \omega^2 - \frac{l(l+1)}{r^2}(1-\frac{2M}{r})\right) f(r^*) = 0 \tag{53}$$

the solution to this will give us the function for the gauge field which can be found from Φ by using

$$A_\mu = \{0, 0, \partial_z\Phi, -\partial_{\bar{z}}\Phi\} \tag{54}$$

1.3.2 A Second Solution

Now suppose $\phi_1 = \phi_2 = \Psi$. Then (41) now becomes

$$(1-\frac{2M}{r})\partial_r(r^2 A_r) + (1+z\bar{z})^2\partial_z\partial_{\bar{z}}\Psi = 0 \tag{55}$$

and adding equations (38) and (39) we get

$$\begin{aligned}
& L\Psi + \frac{2}{r}(1-\frac{2M}{r})\partial_r\Psi - \frac{2}{r}\left(1-\frac{3M}{r}\right)A_r = 0 \\
\Rightarrow \quad & (1-\frac{2M}{r})^{-1}\partial_t^2\Psi - \frac{1}{r^2}\partial_r\left(r^2(1-\frac{2M}{r})\partial_r\Psi\right) - \frac{(1+z\bar{z})^2}{r^2}\partial_z\partial_{\bar{z}}\Psi \\
& + \frac{2}{r}(1-\frac{2M}{r})\partial_r\Psi - \frac{2}{r}\left(1-\frac{3M}{r}\right)A_r = 0 \\
\Rightarrow \quad & (1-\frac{2M}{r})^{-1}\partial_t^2\Psi - \partial_r\left((1-\frac{2M}{r})\partial_r\Psi\right) - \frac{(1+z\bar{z})^2}{r^2}\partial_z\partial_{\bar{z}}\Psi - \frac{2}{r}\left(1-\frac{3M}{r}\right)A_r = 0
\end{aligned} \tag{56}$$

Let $\Psi(t,r,\theta,\varphi) = (1-\frac{2M}{r})\partial_r\tilde{\Phi}(r)Y_l^m(\theta,\varphi)e^{i\omega t}$

$$\begin{aligned}
&-\omega^2\partial_r\tilde{\Phi}(r) - \partial_r\left((1-\frac{2M}{r})\partial_r\left[(1-\frac{2M}{r})\partial_r\tilde{\Phi}(r)\right]\right) \\
&+(1-\frac{2M}{r})\frac{l(l+1)}{r^2}\partial_r\tilde{\Phi}(r) - \frac{2}{r}\left(1-\frac{3M}{r}\right)\frac{A_r e^{-i\omega t}}{Y_l^m} = 0 \\
&\Rightarrow \quad -\partial_r\left[\left((1-\frac{2M}{r})\partial_r\left[(1-\frac{2M}{r})\partial_r\tilde{\Phi}\right]\right) + \omega^2\tilde{\Phi}\right] \\
&+(1-\frac{2M}{r})\frac{l(l+1)}{r^2}\partial_r\tilde{\Phi} - \frac{2}{r}\left(1-\frac{3M}{r}\right)\frac{A_r e^{-i\omega t}}{Y_l^m} = 0 \\
&\Rightarrow \quad -\partial_r\left[\partial_{r^*}^2\tilde{\Phi} + \omega^2\tilde{\Phi}\right] + \partial_r\left[(1-\frac{2M}{r})\frac{l(l+1)}{r^2}\tilde{\Phi}\right] \\
&-\partial_r\left[(1-\frac{2M}{r})\frac{l(l+1)}{r^2}\right]\tilde{\Phi} - \frac{2}{r}\left(1-\frac{3M}{r}\right)\frac{A_r e^{-i\omega t}}{Y_l^m} = 0 \\
&\Rightarrow \quad -\partial_r\left[\partial_{r^*}^2\tilde{\Phi} + \left(\omega^2 - (1-\frac{2M}{r})\frac{l(l+1)}{r^2}\right)\tilde{\Phi}\right] \\
&+\frac{2}{r}\left(1-\frac{3M}{r}\right)\left[\frac{l(l+1)}{r^2}\tilde{\Phi} - \frac{A_r e^{-i\omega t}}{Y_l^m}\right] = 0
\end{aligned} \tag{57}$$

This Is simplified if $A_r(t,r,\theta,\varphi) = l(l+1)\frac{\tilde{\Phi}(r)}{r^2}Y_l^m(\theta,\varphi)e^{i\omega t}$, in which case the equation of motion becomes

$$\begin{aligned}
&\partial_r\left[\partial_{r^*}^2\tilde{\Phi} + \left(\omega^2 - (1-\frac{2M}{r})\frac{l(l+1)}{r^2}\right)\tilde{\Phi}\right] = 0 \\
&\Rightarrow \quad \left(\partial_{r^*}^2 + \omega^2 - \frac{l(l+1)}{r^2}(1-\frac{2M}{r})\right)\tilde{\Phi} = 0
\end{aligned} \tag{58}$$

Let's call $\Phi(t,r,\theta,\varphi) = \tilde{\Phi}(r)Y_l^m(\theta,\varphi)e^{i\omega t}$. The second solution for the EM field is given by.

$$A_\mu = \left\{0, \frac{l(l+1)}{r^2}\Phi, (1-\frac{2M}{r})\partial_z\partial_r\Phi, (1-\frac{2M}{r})\partial_{\bar{z}}\partial_r\Phi\right\} \tag{59}$$